

Problem:

We are given the following system of congruences:

1. $x \equiv 1 \pmod{5}$
2. $x \equiv 2 \pmod{7}$
3. $x \equiv 3 \pmod{9}$
4. $x \equiv 4 \pmod{11}$

Steps:

Step 1: Calculate the product of all moduli (M)

$$M = 5 \cdot 7 \cdot 9 \cdot 11 = 3465$$

Step 2: Compute M_i for each congruence

For each congruence $x \equiv a_i \pmod{n_i}$, calculate:

$$M_i = \frac{M}{n_i}$$

- For $n_1 = 5$: $M_1 = \frac{3465}{5} = 693$
- For $n_2 = 7$: $M_2 = \frac{3465}{7} = 495$
- For $n_3 = 9$: $M_3 = \frac{3465}{9} = 385$
- For $n_4 = 11$: $M_4 = \frac{3465}{11} = 315$

Step 3: Compute modular inverses

Handwritten calculations for finding modular inverses:

M_1
 $693 \cdot y_1 \equiv 1 \pmod{5}$
 $3 \cdot y_1 \equiv 1 \pmod{5}$
 $3 \cdot (2) \equiv 1 \pmod{5} \rightarrow \boxed{6 \pmod{5} = 1}$
 $y_1 = 2$

M_2
 $495 \cdot y_2 \equiv 1 \pmod{7}$
 $5 \cdot y_2 \equiv 1 \pmod{7}$
 $5 \cdot (3) \equiv 1 \pmod{7} \rightarrow \boxed{15 \pmod{7} = 1}$
 $y_2 = 3$

M_3
 $385 \cdot y_3 \equiv 1 \pmod{9}$
 $7 \cdot y_3 \equiv 1 \pmod{9}$
 $7 \cdot (4) \equiv 1 \pmod{9} \rightarrow \boxed{28 \pmod{9} = 1}$
 $y_3 = 4$

$$\begin{array}{l}
 \boxed{14} \quad 315 \cdot y_4 \equiv 1 \pmod{11} \\
 7 \cdot y_4 \equiv 1 \pmod{11} \\
 7 \cdot (8) \equiv 1 \pmod{11} \\
 y_4 = 8
 \end{array}
 \left|
 \begin{array}{l}
 315 \pmod{11} = 7 \\
 58 \pmod{11} = 1
 \end{array}
 \right.$$

$$\begin{aligned}
 x &= (1 \cdot 693 \cdot 2) + (2 \cdot 495 \cdot 3) + (3 \cdot 385 \cdot 4) \\
 &\quad + (4 \cdot 315 \cdot 8) \pmod{3465} \\
 &= 19056 \pmod{3465} \\
 &= 1731
 \end{aligned}$$

The Sieve of Eratosthenes algorithm is an efficient way to find all prime numbers up to a given number n . Here, we will simulate the steps of the algorithm to find all primes not exceeding 50.

Steps of the Sieve of Eratosthenes

Initialization:

Start with a list of integers from 2 to 50. All numbers are initially assumed to be prime.

List: 2, 3, 4, 5, 6, 7, 8, 9, 10,
 11, 12, 13, 14, 15, 16, 17, 18, 19, 20,
 21, 22, 23, 24, 25, 26, 27, 28, 29, 30,
 31, 32, 33, 34, 35, 36, 37, 38, 39, 40,
 41, 42, 43, 44, 45, 46, 47, 48, 49, 50

Steps:

1. Cross out all multiples of 2 (except 2 itself) as they cannot be prime.
2. **Cross out all multiples of 3 (except 3 itself).**
3. **Cross out all multiples of 5 (except 5 itself).**
4. **Cross out all multiples of 7 (except 7 itself).**

Answer:

2, 3, 5, 7, 11, 13, 17, 19, 23, 29, 31, 37, 41, 43, 47

To compute $5^{51} \bmod 100$ efficiently, we can use **modular exponentiation** with optimizations. step-by-step:

We aim to compute $5^{51} \bmod 100$.

Step 1: Break 5^{51} into smaller powers

Using modular arithmetic properties, split 5^{51} into successive powers of 5, reducing at each step:

$$5^{51} = (5^{50} \cdot 5) \bmod 100$$

Step 2: Simplify $5^{50} \bmod 100$

Notice that $5^2 = 25$, and any higher power of 5 will always end in 25. This means:

$$5^n \bmod 100 = 25 \quad \text{for } n \geq 2$$

Thus, $5^{50} \bmod 100 = 25$.

Step 3: Compute $5^{51} \bmod 100$

Now, substitute $5^{50} \bmod 100 = 25$:

$$5^{51} = (5^{50} \cdot 5) \bmod 100 = (25 \cdot 5) \bmod 100$$

Simplify:

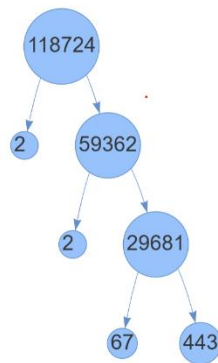
$$25 \cdot 5 = 125 \quad \text{and } 125 \bmod 100 = 25$$

Final Answer:

$$5^{51} \bmod 100 = 25$$

<https://math.tools/numbers/prime-factors>

Prime factors of 118724



$$f(p) \equiv (p + 14) \pmod{26}$$

Step 1: Assign numbers to letters

Using the standard English alphabet:

- A = 0, B = 1, C = 2, ..., Z = 25

Step 2: Convert each letter to a number

For the message "Welcome to IIUC":

- W = 22, E = 4, L = 11, C = 2, O = 14, M = 12, E = 4
- T = 19, O = 14
- I = 8, I = 8, U = 20, C = 2

Step 3: Apply the encryption function

$$f(p) \equiv (p + 14) \pmod{26}$$

Encrypt each number:

1. $f(22) = (22 + 14) \pmod{26} = 10 \rightarrow \text{K}$
2. $f(4) = (4 + 14) \pmod{26} = 18 \rightarrow \text{S}$
3. $f(11) = (11 + 14) \pmod{26} = 25 \rightarrow \text{Z}$
4. $f(2) = (2 + 14) \pmod{26} = 16 \rightarrow \text{Q}$
5. $f(14) = (14 + 14) \pmod{26} = 2 \rightarrow \text{C}$
6. $f(12) = (12 + 14) \pmod{26} = 0 \rightarrow \text{A}$
7. $f(4) = (4 + 14) \pmod{26} = 18 \rightarrow \text{S}$
8. $f(19) = (19 + 14) \pmod{26} = 7 \rightarrow \text{H}$
9. $f(14) = (14 + 14) \pmod{26} = 2 \rightarrow \text{C}$
10. $f(8) = (8 + 14) \pmod{26} = 22 \rightarrow \text{W}$
11. $f(8) = (8 + 14) \pmod{26} = 22 \rightarrow \text{W}$
12. $f(20) = (20 + 14) \pmod{26} = 8 \rightarrow \text{I}$
13. $f(2) = (2 + 14) \pmod{26} = 16 \rightarrow \text{Q}$

The encrypted message is:

"KSZQCASHCWWIQ"

Statement to Prove:

For $n \geq 4$, any amount n can be paid using a combination of 2-taka and 5-taka notes.

Step 1: Base Cases

Check for $n = 4$ and $n = 5$:

1. $n = 4$: Use two 2-taka notes: $2 + 2 = 4$.
2. $n = 5$: Use one 5-taka note: $5 = 5$.

Thus, the base cases are true.

Step 2: Induction Hypothesis

Assume that for some $k \geq 4$, any amount k can be paid using a combination of 2-taka and 5-taka notes.

That is, $k = 2a + 5b$ for some non-negative integers a and b .

Step 3: Induction Step

We need to prove that $k + 1$ can also be paid using a combination of 2-taka and 5-taka notes.

Case 1: If $a \geq 1$

- Reduce one 2-taka note from k (i.e., $a - 1$).
- Add a new 2-taka note to reach $k + 1$:

$$k + 1 = 2(a - 1) + 2 + 5b = 2a + 5b + 1$$

This means $k + 1$ can still be paid using a valid combination of notes.